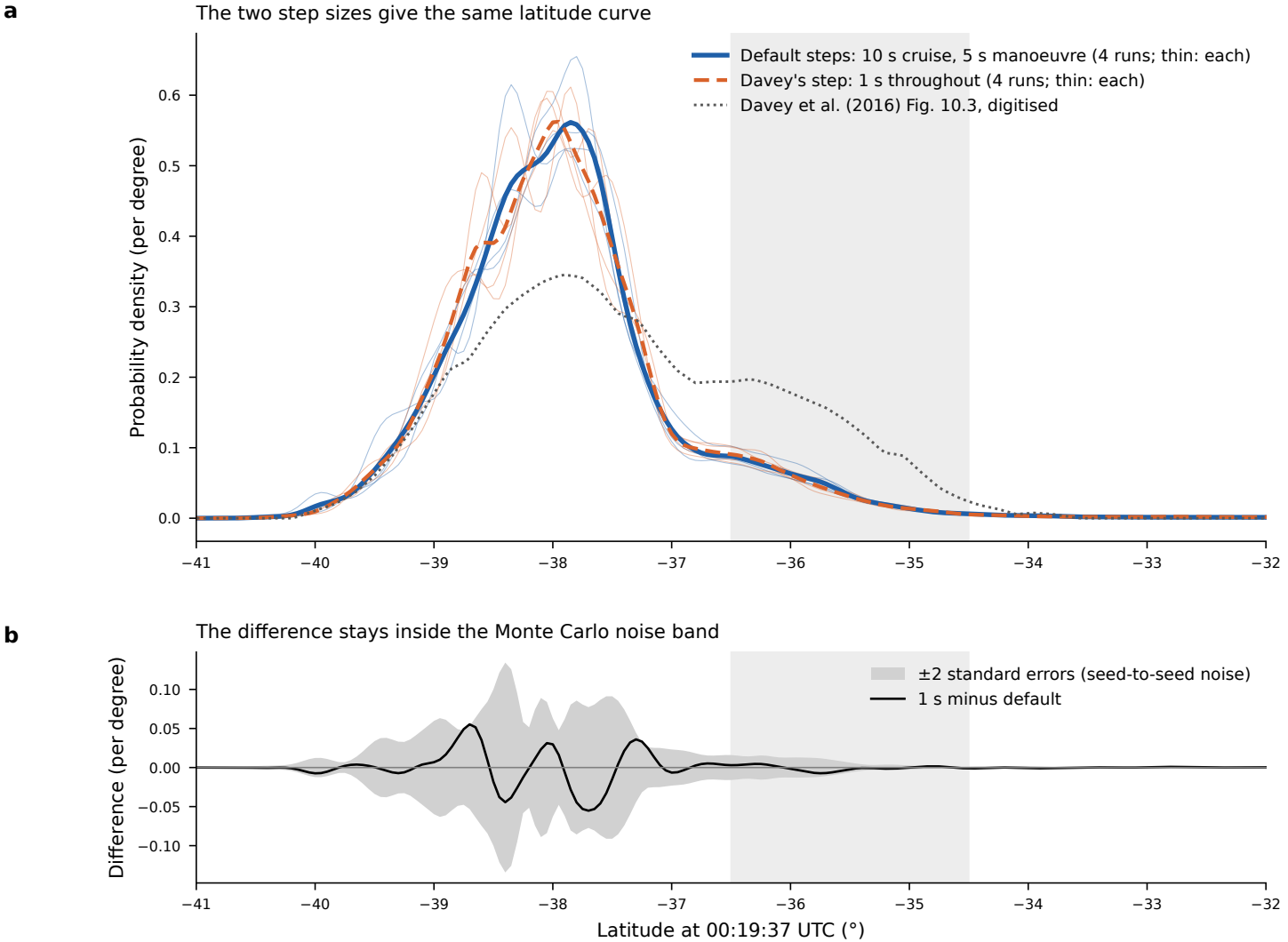


Plain (ASD-STE100): This is the step-size test of 15 October 2026. We flew Davey's model alone two times: with our default time steps (10 s in cruise, 5 s in turns and speed changes) and with Davey's 1 s step. The two curves are almost the same. Their difference (panel b) stays inside the range that random sampling alone gives. The grey band is the published northern shoulder, 34.5–36.5° S. Thus the step size does not change the result, and it does not explain why our shoulder is smaller than Davey's. The test used Davey's model only, with BFO noise of 7 Hz, not the full run-J model.

Technical: audit F3, pure reproduction config/davey2016-inmarsat.toml (no fuel, speed envelope, radar or families), BTO+BFO, BFO sd 7 Hz, 4 seeds × 7M particles, deskstar 88 threads. 1 s: job cae37735, bin-b857425, step-check/repro-step1.toml (46 min). Default: job ac952234, bfo-sd-check/repro-bfo7.toml, bin-dbeb59e (8 min). Same seeds; not byte-comparable. ln Z −97.163 vs −97.128 (z = −1.3 from seed spread); median -37.968 vs -37.952 (engine summary); mean latitude -37.736 vs -37.729 (difference -0.007 ± 0.036°); shoulder 0.0769 vs 0.0783; PDF overlap 0.966; split-half 0.945 / 0.936. Panel b: pooled-density difference with ±2 SE from the per-seed variance of both arms (4% of 40–33° S grid points outside, consistent with noise). Weighted particle histograms, 0.1° Gaussian smoothing. results/step-check-2026-10-15/. Built 18 Oct 2026.